University of Rajshahi
(For Affiliated Colleges)
Department of Computer Science and Engineering
B.Sc. Engineering Part-1 Even Semester Examination 2021
Course: MATH1221 (Co-Ordinate Geometry, Vector Analysis and Complex Variables)
Marks: 52.5 Time: 3 Hours
Session:2019-2020

(Answer SIX questions taking at least THREE from each Section)

Section-A

1.(a) Find the transformation equation to change the origin without changing the direction of the axes. 3
(b) Transform to axes inclined at 30^0 to the original axes of the equation $x^2 + 2\sqrt{3}xy - y^2 = 2a^2$. 3
(c) Find the ratio in which the line segment joining the points (2, -3, 5) and (7, 1, 3) is divided by the XY-plane. 2.75

2. (a) Rotate axes to eliminate the xy-term from the equation $ax^2 + 2hxy + by^2 = c$. 3
(b) A, B, C are three points on the axis of x, y and z respectively at distances a, b, c from the origin. Find the co-ordinate of the point which is equidistance from A, B, C and O. 2.75
(c) Prove that the angle between two diagonals of a cube is $\cos^{-1}\frac{1}{3}$. 3

3. (a) A variable plane is at a constant distance p from the origin and meets the axes in A, B and C. Show that the locus of the centroid of the tetrahedron $OABC$ is $x^{-2} + y^{-2} + z^{-2} = 16p^{-2}$. 2.75
(b) Find the symmetrical form of the equation of the line $x + y + z + 1 = 0$, 3
(c) $4x + y - 2z + 2 = 0$ and find its direction-cosines. 3
Find the shortest distance between the lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$.

4. (a) Evaluate $\int_c \bar{F}.d\bar{r}$ where $\bar{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$, curve C is the rectangle in xy-plane bounded by $y = 0, x = a, y = b, x = 0$. 3
(b) Show that $\nabla\emptyset$ is a vector perpendicular to the surface $\emptyset(x, y, z) = c$, where c is a constant. 3
(c) If $\vec{v} = \vec{w} \times \vec{r}$, prove that $\vec{w} = \frac{1}{2} curl\ \vec{v}$, where $\vec{w} = w_1\hat{i} + w_2\hat{j} + w_3\hat{k}$ is a constant vector. 2.75

Section-B

5. (a) A particle moves along the curve $x = 2t^2,\ y = t^2 - 4t,\ z = 3t - 5$, where t is the time. Find the components of its velocity and acceleration at time t=1 in the direction $\hat{i} - 3\hat{j} + 2\hat{k}$. 4
(b) Show that $\vec{A}.\frac{d\vec{A}}{dt} = A\frac{dA}{dt}$. 2.75
(c) Find $\vec{\nabla}\emptyset$ if $\emptyset = \frac{1}{\vec{r}}$ 2

6. (a) Show that $\frac{d\bar{z}}{dz}$ does not exist anywhere. 3
(b) Show that the function $f(z) = x^2 - y^2 + i2xy$ is differentiable everywhere. 3
(c) Evaluate $\int_c (x + y)dx + x^2ydy$ along $y = 3x$ between the points (0, 0) , (3, 9). 2.75

7. (a) State Cauchy integral formula. Evaluate $\frac{1}{2\pi i}\oint\frac{e^z}{z-2}dz$ if c is (i) the circle $|z| = 3$, (ii) the circle $|z| = 1$. 6
(b) Evaluate $\int_c \frac{e^{2z}}{(z-1)(z-4)}dz$ where c is the circle $|z| = 2$. 2.75

8. (a) Define analytic function at a point. If $w = f(z) = \frac{1+z}{1-z}$, find (i) $\frac{dw}{dz}$, (ii) determine where $f(z)$ is non-analytic. 4
(b) Define residue at a point. Find the residues of $f(z) = \frac{z}{(z-1)(z+1)^2}$ at all its poles in the finite plane. 4.75

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Department of Computer Science and Engineering
B.Sc. Engineering Part-1 Even Semester Examination 2021
Course: MATH1221 (Co-Ordinate Geometry, Vector Analysis and Complex Variables)
Marks: 60 Time: 3 Hours
Session:2020-2021

(Answer SIX questions taking at least THREE from each Section)

Section-A

1.(a) Find the transformation equation to change the directions of the axes without changing the origin, both pairs of axes are rectangular. 4

(b) Transform to axes inclined of 30^0 to the original axes of the equation $x^2 + 2\sqrt{3}xy - y^2 = 2a^2$. 3

(c) Transform the axes through the point (2, 2) and rotating the axes through an angle 45^0 to the original of the equation $3x^2 + 3y^2 + 2xy - 2 = 0$. 3

2. (a) Define direction Cosines and direction ratios. Find the direction cosines of the line passing through the two points (-2, 4, -5) and (1, 2, 3). 4

(b) Find the equation of the plane passing through the points (1, -2, 2) and (-3, 1, -2) and perpendicular to the plane $2x + y - z + 6 = 0$. 3

(c) Find the equation of the straight line passing through (2, -1, 1) and parallel to the line joining the points (1, 2, 3) and (-1, 1, 2). 3

3. (a) Find an equation for the plane perpendicular to the vector $\bar{A} = 2\hat{\imath} + 3\hat{\jmath} + 6\hat{k}$ and passing through the terminal point of the vector $\bar{B} = \hat{\imath} + 5\hat{\jmath} + 3\hat{k}$. 4

(b) Find an equation for the plane determined by the points $P_1(2, -1, 1)$, $P_2(3, 2, 1)$ and $P_3(-1, 3, 2)$. 3

(c) A particle moves along the curve $x = 3t^2$, $y = t^3 - 4t$, $z = 2t$, where t is the time. Find the components of the velocity and acceleration at time $t = 2$ in the direction $\mathrm{i} - 4\mathrm{j}$. 3

4. (a) Find the directional derivative of $\emptyset = x^2yz + 4xz^2$ at (1, -2, -1) in the direction $2\hat{\imath} - \hat{\jmath} - 2\hat{k}$. 3

(b) If $\vec{v} = \vec{w} \times \vec{r}$, prove that $\vec{w} = \frac{1}{2} curl\, \vec{v}$, where $\vec{w}$ is a constant vector. 3

(c) Find the total work done in moving a particle in a force field given by $\vec{F} = 3xy\hat{\imath} - 5z\hat{\jmath} + 10x\hat{k}$ along the curve $x = t^2 + 1, y = 2t^2, z = t^3$ from $t = 1$ to $t = 2$. 4

Section-B

5. (a) Define complex number. Discuss the graphical representation of complex number. 4

(b) Define limit of a complex function. Show that if $f(z) = \frac{z}{\bar{z}}$ then $\lim_{z \to 0} f(z)$ does not exist. 3

(c) Show that $\frac{d\bar{z}}{dz}$ does not exist anywhere. 3

6. (a) State Cauchy-Riemann equation. Show that the function $f(z) = z^2$ is differentiable. 4

(b) Express the Cauchy-Riemann equation in polar form. 3

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B.Sc. Engg. Part-I, Even Semester, Examination 2020

Course Code: MATH-1221

Course Title: Co-ordinate Geometry, Vector Analysis and Complex Variable

Time: 3 Hours **Full Marks: 52.5**

[N.B. Answer SIX questions taking at least THREE from each Section.]

Section-A

1. (a) Construct the rotation matrix when the point is rotated through an angle θ. 3.75

   (b) If the point is rotated through $45°$, find the new coordinates of the point whose coordinates are $(1,1)$ using the rotation matrix. 3

   (c) Prove that rotation is distance invariant. 2

2. (a) Rotate axes to eliminate the xy-term from the equation $7x^2 - 6\sqrt{3}xy + 13y^2 - 16 = 0$ and draw the figure. 3

   (b) What surfaces in $\mathbb{R}^3$ are represented by the following equations? 1.75
   $(i)\ x = 5 \quad (ii)\ z = 4.$

   (c) A line makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube; show that 4
   $\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = \frac{4}{3}$.

3. (a) Prove that the general equation of first degree in x, y, z *i.e.*, 3
   $Ax + By + Cz + D = 0$ represents a plane.

   (b) Find the equation of the straight lines through $(2, -1, 4)$, which are 2.75
   (i) parallel to y-axis (ii) perpendicular to y-axis.

   (c) Find the shortest distance between the lines 3
   $$\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}; \qquad \frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}.$$

4. (a) Forces $\overrightarrow{F_1}, \overrightarrow{F_2}, \overrightarrow{F_3}, \overrightarrow{F_4}, \overrightarrow{F_5}, \overrightarrow{F_6}$ act as shown on object P. What force is needed to prevent P from moving? 1.75

